

DRY-RUN GUIDE | sparse_matrix.py

Source: Python\3rd_Belt\Matrix\sparse_matrix.py

CORE BEHAVIOR

Function-only module: returns True when at least half of an M-by-N matrix's elements are zero.

REPRESENTATIVE INPUT

Call `is_sparse([[0,1],[0,0]],2,2)`.

STEP TRACE

Three zeros; four cells total; $3 \geq 4/2$ is True.

EXPECTED OUTPUT / RETURN

Returns 'True'; import itself prints nothing.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\Matrix\sparse_matrix.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Check whether a matrix is sparse by counting how many zero elements it contains.
002 | # Add your solution here.
003 |
004 | def is_sparse(matrix, M, N):
005 |     zero_count = sum(row.count(0) for row in matrix)
006 |
007 |     total_elements = M * N
008 |
009 |     return zero_count >= (total_elements / 2)
010 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.